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Personalized Multi-document Text Summarization using Deep Learning Techniques

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Abstract

The summary of a large piece of text or a document is a concise description of the same thing. It must retain most of the important points from the document and remove any kind of verbosity. The text summarization task is an information processing scheme that given a document or a set of documents, mines the most essential content from the source by considering the user or task at hand and presents the summary in a well-formed and brief text. Summarization can be done not just on one document, it can be done on a set of documents as well, which is called multi-document summarization. The proposed approach considers and in-corporates user preferences for the summarization task. Depending on the current user or current task, the summary could differ essentially. For an instance, a summary on ‘photosynthesis’, would be very different for the kid in 4th Grade vs. a student doing a bachelor’s in Science. Therefore, while writing a summary, the reading history of the user is employed as a source of personalizing text summarization tasks. The proposed method is experimentally evaluated in the domain of news articles and obtained better summaries capable of extracting important concepts based on user preferences explained in the document when considering the relevant domain terms in the process of multi-document text summarization.

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1. Introduction

People generally see the headline of the news, after all a summary. The news snippet is also a reasonable summary of the entire news article. The News system makes use of the summarization system the most. Everyone wants to keep themselves abreast of all the news which is going on. However, we do not have time i.e. nobody has time to read the entire newspaper from start to end. Therefore, we generally install news readers on our Phones. This news reader application helps us by presenting headlines or essentially say a summary of the entire news article. Now, those news readers, vary in their complexity of summarization. Therefore, some news readers just give us the 1st paragraph of the entire document and call that a summary, while some news readers are intelligent that they pick up very interesting sentences from the entire document and present a summary or gist of the document [1]. Therefore, this paper focuses on creating an extractive summary out of multiple news documents.

The other scenario is that researchers and scientists generally publish various articles. Various research institutions publish multiple research papers. If we want to read the published papers, we cannot read all the papers published in the world, even in our own areas of interest. If we are interested in data science, and if we want to read all data science papers, it is impossible to read, because, there are many papers that come up in data science every day. One way is to read only the abstract of those papers. Every paper comes with an abstract, which is usually like a short summary. It is made up of 100 to 240 words of summary of the entire paper. A good paper usually has the abstract as the summary of the introduction. Therefore, the 1st section is usually the introduction after the abstract; for instance, an introduction is the summary of the entire research paper. A reader can think of a research article as four-level content. There is a heading which is essentially a summary of the abstract; an abstract, which is like the summary of the introduction; an Introduction is like the summary of the entire paper. Therefore, for different kinds of users, different help of summary helps. In fact, by looking at the title, the reader can decide whether he/she wants to read the paper or not. The reader then looks at the abstract section, then decides whether to go detail into about it or not. In general, the reader read the abstracts of all papers. However, the reader read papers in detail, only if they are of his/her interests or preferences. Essentially, text summarization saves time a lot when we want to figure out papers of our interest, then only we read important parts of other papers, which are not of interest. The rest of the paper is organized as follows; the related works and background are presented in Section 2. The proposed personalized summarization model is presented in Section 3. Section 4 illustrates the dataset, baseline models, and evaluation metrics considered for empirical evaluation. The experimental results are described in Sections 4.4 and 4.5. Finally, section 5 concludes the paper with future directions.

Nomenclature

U	user preference
C	output summary
L	similarity between user preference and the output summary
y	a binary indicator denoting whether the two text segments belong to the same category or not

2. Related Works

There are different types of summarization techniques proposed in the literature, which have been discussed as follows.

2.1. Single-document vs. multi-document summarization

Single document summarization is about generating a summary out of a single document, whereas multi-document summarization generates a summary of the news event by aggregating information from thousands of news articles. The multiple documents come from multiple sources, and somehow combine the information from all the sources to generate a summary. That is called multi-document text summarization. In this paper, a multi-document summarization technique has been proposed and analyzed.

2.2. Generic vs. Query-based vs. User-focused

In a generic summary, all major topics must be covered with almost equal coverage. For instance, assume that someone has given us a book on World War I and asked us to write a summary of that book. Query-based summary [2] where the idea is that a book document will be given and then a summary will be written on the topic what are the causes of World War-I. Here the idea is that the automated summarization system must be able to figure out the parts of the book which are the relevant to causes of the War. This is called query-based summarization. If we ask someone to write a summary for a kid who is in 4th Grade or to write a summary for a historian who has been doing research in History over like 10 years, a summary could be written again a historian guy or some guy who is Ph.D. in history. Now, for those kinds of readers, we need to understand what kind of background they have and then come up with summaries. For a historian, very complicated stuff could be written and skip up a few links as well and the historian will be able to understand. However, a Ph.D. student in Chemistry or a 4th Grade kid, those guys are not really as great at history. Therefore, they will not be able to write or understand summaries which are advanced summaries in some senses.

2.3. Extractive vs. abstractive summarization

Extractive summarization is about extracting sentences and then putting them out as summaries, whereas abstractive summarization is that understanding the stuff and writing in our own words. An automated summarization system could be extractive or abstractive in nature. Abstractive systems [3, 4] are mostly in research. They do not work as well as expected. Extractive systems are still better. However, extractive systems have the problem of coming up with sentences that may not be nicely linked with each other. And therefore, the readability of the entire passage may not be very good. But, it turns out they are still better than abstractive ones because for abstractive ones; coming up with correct grammar sentences is itself a challenge right now. Therefore, people usually end up doing an extractive kind of summarization. Following are the techniques of extractive summarization; position-based approach, cue phrase-based approach, word frequency-based method, and information extraction-based approach.

2.4. Bidirectional Encoder Representations

Training a language model on a very large corpus requires extensive computation. Therefore, pre-trained language models [5] have been exploited for the task of both extractive and abstractive text summarization [6, 7, 8]. The document-level bidirectional encoder model has the capability to precisely articulate the semantics of a text document and get semantic representations for its sentences.

2.5. GameWikiSum

It is an approach that uses a comprehensive video game review dataset for multi-document summarization. It is a dataset exclusively prepared for the large-scale multi-document summarization task, even though most of the available dataset for text summarization is only in the news domain [9].

2.6. Informative summaries

Informative summaries include quantitative and qualitative information. They basically talk more about an event. For an instance, in the case of an event of an earthquake, they generate a summary about where it happened, what is the Richter scale value, when did it happen, etc.

3. Proposed Model

3.1. Problem Statement

The problem which is addressed in this paper is that summary generation by extracting salient sentences from multiple documents. Upon taking multiple documents as input, a summary of top k non-redundant salient sentences is going to be generated by sentence ranking and sentence extraction. The dataset used in this work is Daily Mail corpus with 200K news articles [10].

3.2. Text preprocessing

Building a model on disorganized data may often lead to poor learning from it despite its volume. Therefore, a few preprocessing or data preparation techniques are employed before the data is fed to the model for training. It is attempted to remove unnecessary white spaces, stop words and unreadable special symbols. The contraction mapping is done to avoid contractions in the summary text. The text data is further processed in all lower case form, except some standard acronyms.

3.3. Long Short Term Memory (LSTM) and Recurrent Neural Network (RNN) for Multi-document Extractive summarization

Text summarization is also a natural language processing task [11] where the idea is that there are multiple documents and to output, a summary of top k non-redundant salient sentences i.e. important sentences must be part of the summary. The goal is to do sentence ranking and sentence extraction. Across these multiple documents; it tries to find out significant sentences to generate a summary based on these sentences. These are the sentences carrying importance scores for each sentence and therefore, take the sentences which have high importance scores and output them as summaries [12, 13, 14]. The overall architecture of the proposed solution is presented in Fig. 1.

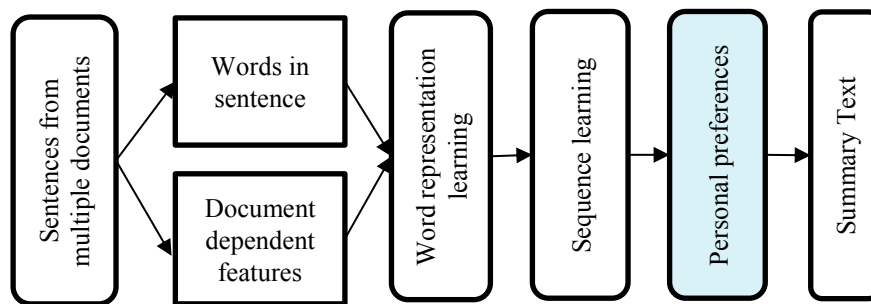


Fig. 1. The overall architecture of the proposed RNN LSTM solution.

- *Word representation learning.* This helps in representing words as feature vectors to make it ready for learning from data. Every entry in feature vector corresponds to a hidden feature from the word meaning. The feature vector essentially represents the semantic dependencies among the words present in the document. Word2Vec is the widely used word representation that uses the skip-gram model. This represents the word feature vector in terms of its neighbors. This attempts to predict all the surrounding words of a given word in the defined context. Another popular distributed word representation is Global Vectors (GloVe). The *Word2Vec* model generally captures some local context, whereas the GloVe embedding uses the overall co-occurrence statistics of words from the document corpus. Since GloVe is computationally expensive, the Word2Vec model is employed in the proposed work that captures whether the words appear together in similar contexts.
- *Sequence learning.* It is the task of mapping text segments from multiple documents of long length into a text summary of a small length in terms of the number of words in it. In this respect, Sequence-to-sequence (seq2seq)

model solves the task as follows; upon taking the given input documents, the seq2seq model generates an encoded representation, which is then used to generate the desired text summary output.

3.4. Personalized Recursive Neural Network (ReNN) architecture for summarization

A technique that uses a Recursive Neural Network (ReNN) has been proposed. The ReNN is like a RNN. However, it has a hierarchy in it. This approach takes a sentence and takes each word in a sentence. For every word, get the Embedding i.e. get the *Word-to-Vec* Embeddings using Word-to-Vec GloVe embedding. That gives a matrix per sentence. The matrix is more like an image. Therefore, the Convolutional Neural Network (CNN) has been used on top of it. Thus, CNN is used per sentence and obtained as a representation for every sentence. For each of those matrices, get the representation. Now, for each of those representations; form the tree-like hierarchy.

This tree is basically the hierarchical structure that can be imposed on a ReNN so that the lowest layer is a ReNN by itself and then, combine the output and form a ReNN at the higher level. The combination of these different levels of recurrent neural networks is called a recursive neural network. Overall, ReNN is combined with CNNs in order to incorporate the user preferences of the reader for generating the summary text. The Siamese Network (SN) [15] is employed to find the closeness between user preference and candidate summary. The SN contains two identical sub-neural networks with the same configuration and weights are shared between them. This helps in finding the similarity between user representation and candidate representation by comparing their feature vectors.

$$L = (U_{\text{preference}}, C_{\text{summary}}, y) \quad (1)$$

Eq. (1) is used to approximate the closeness or similarity L between user preference $U_{\text{preference}}$ and the output summary C_{summary} . The notation y is a binary indicator denoting whether the two text segments belong to the same category or not.

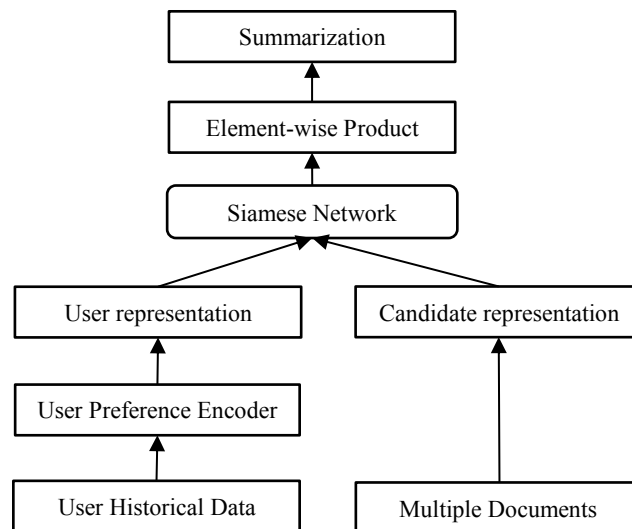


Fig. 2. Workflow of incorporating user preferences into text summarization task.

In this approach, temporal aspect of user preference is incorporated which is completely missing in most of the related works. Therefore, it turns out that the sequence in which the user read this articles is also important. If *user* is reading an article, ‘*which article user will read next*’ also depends on the sequence in which the *user* has read the previous articles. The *user article reading history* has been modeled as a sequence of articles [16, 17]. Therefore, it is basically a sequence in which the users read those articles. The goal here is to summarize the set of news articles according to the user U . The idea was to basically use the user’s historical data. Use a ReNN based or an LSTM-

based encoder so as to encode the information from the user's historical data. And, combine it with the candidate representation or the news article representation. Then, check whichever has a very good similarity with the user's reading history and is the one that is used to generate the summary. The user's reading history has been taken for building the user preference model. Assume that the user read the articles $r_1, r_2, r_3 \dots r_t$ up to time period t . We basically have these hidden layers for an LSTM. We basically have two kinds of LSTMs. One is the usual one and the other one is where we also have attention i.e. Attentional LSTM as it is called. We took both user model and candidate articles and combined their outputs and then passed them through more layers or Multilayered Perceptron. Finally, we get the summary of multiple news articles. If it does not generate the proper summary, backpropagate and update all the weights all the way from the Multilayered Perceptron layers to RNN layers.

4. Empirical evaluation

4.1. Dataset descriptions

The proposed work has been evaluated using two different datasets as shown below.

- *Daily Mail dataset* [<https://cs.nyu.edu/~kcho/DMQA/>] comprises the text documents from the Daily Mail news articles. There are roughly 200K documents in this repository.
- *MultiNews dataset* comprises news articles and human-transcribed precise summaries. The dataset originates from a varied set of news bases from over 1500 news websites.

4.2. Baseline Models

Text Rank. It is a graph-based ranking model for text summarization. The successful applications of graph theoretic concepts including Hyperlink Induced Topic Search (HITS) and Page Rank are being the motivation behind this approach [18]. The Text Rank model helps in extracting keywords and sentences for building the summaries in an unsupervised manner by deciding the importance of a vertex in the text graph representation.

Standard CNN. The document embedding is fed into the standard convolution layer with ReLU as its activation. The Max pool layer followed by the dropout layer is used. The fully connected (FC) layers use sigmoid activation and the cross entropy loss function is employed to estimate the error [19].

HNet. This model takes each word in a sentence. For every word, get the *Word2Vec* embedding, which provides a matrix per sentence. This matrix is more like an image. Therefore, it uses CNN on top of it. It employs CNN per sentence and gets the representation for every sentence. Each of these representations forms a node of the tree called the convolutional sentence tree indexer. This tree is a hierarchical structure that is imposed on an RNN [20].

4.3. Evaluation Metrics Used

The *Recall-Oriented Understudy for Gisting Evaluation* (ROUGE) score has been used for evaluating the proposed summarization technique. The following variant of the ROUGH metric [21, 22] has been measured in order to substantiate the proposed deep learning-based text summarization.

- *ROUGE-1 score*: this measures the amount of overlap of unigram between the reference summary of the test instance and the generated output summary.
- *ROUGE-2 score*: this measures the amount of overlap of bigram between the reference summary of the test instance and the generated output summary.
- *ROUGH-L score*: this measures the length of the longest common subsequence found in both the reference summary of the test instance and the generated output summary. This finds the common longest co-occurring n-gram sequence in order to figure out the similarity of the test instance and the generated summary.

4.4. Evaluation of LSTM and ReNN model

There are different baseline models for summarization techniques have been proposed over time. The experimental results have been tabulated in Table 1 and Table 2 and it is perceived that the proposed LSTM and ReNN-based technique outperforms the various classical techniques for text summarization.

Table 1. Accuracy scores over Daily mail dataset.

Summarization Models	ROUGH-1	ROUGH-2
LexRank	32.43	5.88
Standard CNN	34.19	7.23
HNet	37.21	8.96
LSTM and ReNN	41.34	9.51

Table 2. Accuracy scores over MultiNews dataset.

Summarization Models	ROUGH-1	ROUGH-2
LexRank	35.29	7.54
Standard CNN	35.73	8.69
HNet	39.17	9.61
LSTM and ReNN	43.94	11.29

It is observed that the deep learning approaches including standard CNN, HNet and LSTM-based summarization techniques observe better accuracy in terms of the *ROUGH scores* compared to the typical LexRank-based model. Among the various deep learning models, the proposed LSTM and ReNN-based model outperforms other models.

4.5. Evaluation of Personalized Model

Dataset. To evaluate the personalized multi-document summarization, the news articles have been crawled from the standard search engine [For this experiment, articles have been crawled from Google] by using the news article headings as input queries. For each query, 15 results are collected, which are often among the top results. For this purpose, only the text portion of the news article has been extracted in order to use as an input source for the summarization task. Along with this, 100 users' search histories have been considered for evaluation of the proposed model. In the data preparation process mentioned above, the reasonable diversity of data is ensured as no benchmark dataset is available for evaluating the personalized summarization model.

Human Evaluation. Besides the objective evaluation discussed above, the performance of the proposed approach has been evaluated through subjective human assessment as well. Randomly selected 50 test examples from the MultiNews test set are collected; and, asked by three annotators who are proficient in English script writing in order to evaluate the outputs of different models individually.

Annotators assess the overall worth of summaries by ranking them by estimating *informativeness* (The final summary carries significant facts of the input articles), *fluency* (The final summary is fluent and grammatically accurate), and *succinctness* (The final summary circumvents repeating information).

Annotators were asked to rank the summaries generated by all the techniques on a scale from 1 to 5 where 1 is considered 'best' and 5 is considered 'worst'. The order of ranking position could be the same for different techniques if they have analogous quality. The rating of each technique is computed as an average of scores on all test examples considered for experimentation.

Table 3. Ranking results of summaries according to human assessment.

Summarization Models	Annotator 1	Annotator 2	Annotator 3	Average score
LexRank	27	25	28	26.67
Standard CNN	31	30	30	30.34
HNet	35	37	38	36.67
LSTM and ReNN	39	41	41	40.34

The proposed LSTM and ReNN-based techniques generate good quality summaries for about 40 test instances as per the rankings prepared by three annotators.

5. Conclusions

In this paper, the importance of personalization in text summarization has been explored using deep learning techniques to improve the quality of the summary that gets generated. The proposed model is able to combine our model with pre-trained LSTMs and RNNs, which further improves the performance, of the multi-document summarization task. A method to incorporate user preferences into the document encoding process has been presented to capture richer relations to generate more informative, fluent, and concise summaries. Experimental results show that the proposed personalized LSTM and RNN-based model outperforms several strong baselines by a wide margin. In the future, other informative graph representation models such as knowledge graphs would be explored and apply them to further improve the quality of the generated summary.

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References

- [1] Alexander R Fabbri, Irene Li, Tianwei She, Suyi Li, Dragomir R Radev. (2019) “Multi-news: a large scale multi-document summarization dataset and abstractive hierarchical model.” In Proc. of the 57th Annual Meeting of the Association for Computational Linguistics, 1074–1084.
- [2] Tal Baumel, Matan Eyal, Michael Elhadad. (2018) “Query focused abstractive summarization: Incorporating query relevance, multi-document coverage, and summary length constraints into seq2seq models.” arXiv.
- [3] Wei Li, Xinyan Xiao, Jiachen Liu, Hua Wu, Haifeng Wang, Junping Du. (2020) “Leveraging Graph to Improve Abstractive Multi-Document Summarization.” arXiv.
- [4] Wei Li. (2015) “Abstractive multi-document summarization with semantic information extraction.” In Proc. of the Conference on Empirical Methods in Natural Language Processing, 1908–1913.
- [5] Yu Sun, Shuohuan Wang, Yukun Li, Shikun Feng, Hao Tian, Hua Wu, Haifeng Wang. (2019) “ERNIE 2.0: A continual pre-training framework for language understanding.” In Proc. of the 33rd AAAI Conference on Artificial Intelligence.
- [6] Yang Liu, Mirella Lapata. (2019) “Text summarization with pre-trained encoders.” In Proc. Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, 3728–3738.
- [7] Joshi A, Fidalgo E, Alegre E, Fernández-Robles L. (2019) “Summocoder. An unsupervised framework for extractive text summarization based on deep auto-encoders.” Expert Systems with Applications, 129: 200–215.
- [8] Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova. (2019) “BERT: Pre-training of deep bidirectional transformers for language understanding.” In Proc. Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, 1: 4171–4186.
- [9] Diego Antognini, Boi Faltings. (2020) “GameWikiSum: a Novel Large Multi-Document Summarization Dataset.” In Proc. 12th Conference on Language Resources and Evaluation, 6645–6650. <https://github.com/Diego999/GameWikiSum> [Last accessed on 30 June 2021].
- [10] Hermann, K. M., Kocisky, T., Grefenstette, E., Espeholt, L., Kay, W., Suleyman, M., Blunsom, P (2015). “Teaching machines to read and comprehend”. In Advances in Neural Information Processing Systems, 1684–1692.

- [11] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. (2014) “Neural machine translation by jointly learning to align and translate.” arXiv preprint arXiv:1409.0473.
- [12] Yadav, A.K., Singh, A., Dhiman, M. (2022) “Extractive text summarization using deep learning approach.” *International Journal of Information Technology*, 14: 2407–2415.
- [13] Al-Maleh M, Desouki S. (2020) “Arabic text summarization using deep learning approach. *Journal of Big Data*.” 7(1):1–17.
- [14] Yang M, Li C, Shen Y, Wu Q, Zhao Z, Chen X. (2020) “Hierarchical human-like deep neural networks for abstractive text summarization.” *IEEE Transactions on Neural Networks and Learning Systems*, 31(6): 2744 - 2757.
- [15] Sounak Dey, Anjan Dutta, J. Ignacio Toledo, Suman K. Ghosh, Josep Lladós, Umapada Pal. (2017) “SigNet: Convolutional Siamese Network for Writer Independent Offline Signature Verification.” *Pattern Recognition Letters*.
- [16] Liu S, Zhou MX, Pan S, Song Y, Qian W, Cai W, Lian X, Tiara. (2012) “Interactive, topic based visual text summarization and analysis.” *ACM Transactions on Intelligent Systems and Technology (TIST)* 3(2):1–28.
- [17] Chen K-Y, Liu S-H, Chen B, Wang H-M, Jan E-E, Hsu W-L, Chen H-H. (2015) “Extractive broadcast news summarization leveraging recurrent neural network language modeling techniques.” *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 23(8):1322–1334.
- [18] Rada Mihalcea and Paul Tarau. TextRANK: Bringing order into text. In *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing*, pp. 404–411, 2004.
- [19] Leo Laugier, Evan Thompson, Alexandros Vlissidis. (2018) “Extractive Document Summarization Using Convolutional Neural Networks – Reimplementation.”
- [20] Singh, Abhishek Kumar and Manish Gupta. (2018) “Neural Approaches towards Text Summarization.”
- [21] Lin, C. Y. (2004) “ROUGE: A package for automatic evaluation of summaries.” In *Proc. Workshop on Text Summarization Branches out (WAS)*, In *Proc. of the ACL-04 workshop*, vol. 8.
- [22] Chin-Yew Lin and Franz Josef Och. (2004) “Automatic Evaluation of Machine Translation Quality Using Longest Common Subsequence and Skip-Bigram Statistics.” In *Proc. 42nd Annual Meeting of the Association for Computational Linguistics*, 605–612.